

Functional genomics using single-cell sequencing

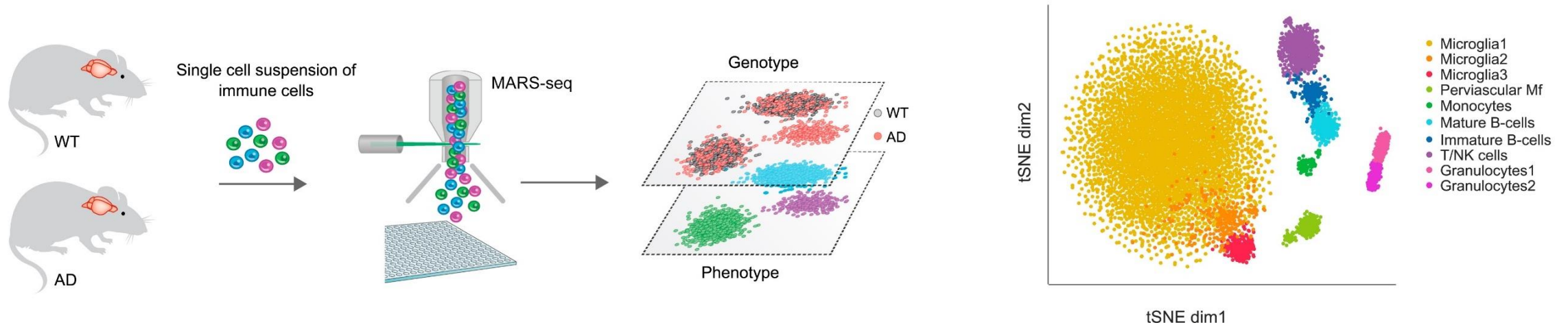
Ahmed Mahfouz

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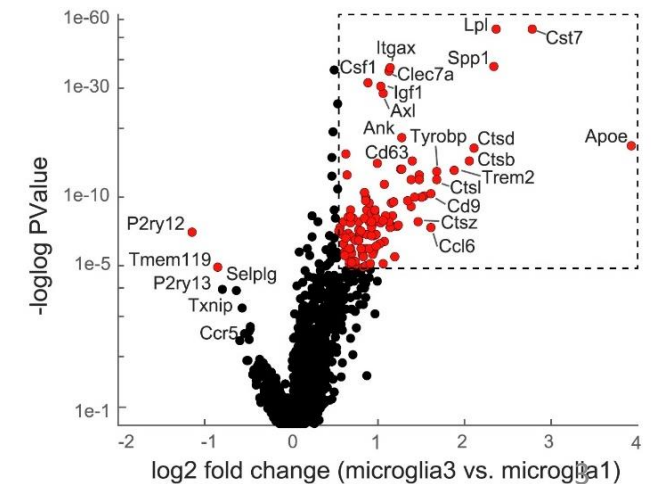
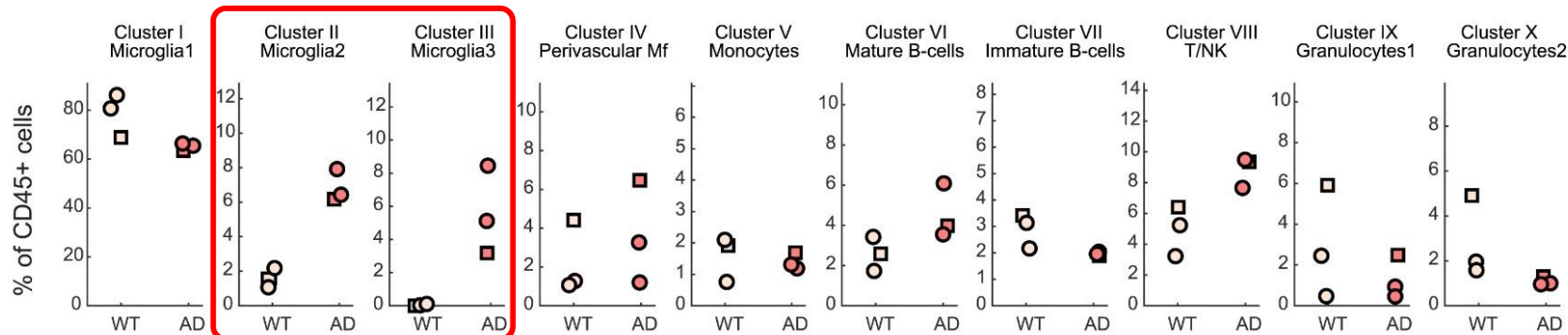
How can single-cell sequencing help functional genomics?

1. Map cell-type specific molecular changes in brain diseases (healthy vs diseased)
2. Identify cell-type specificity of diseases variants (using healthy scRNA-seq, scATAC-seq or)
3. Expression / co-expression quantitative trait loci (eQTL)
4. Perturb-seq

1. Cell-type specific changes in diseases

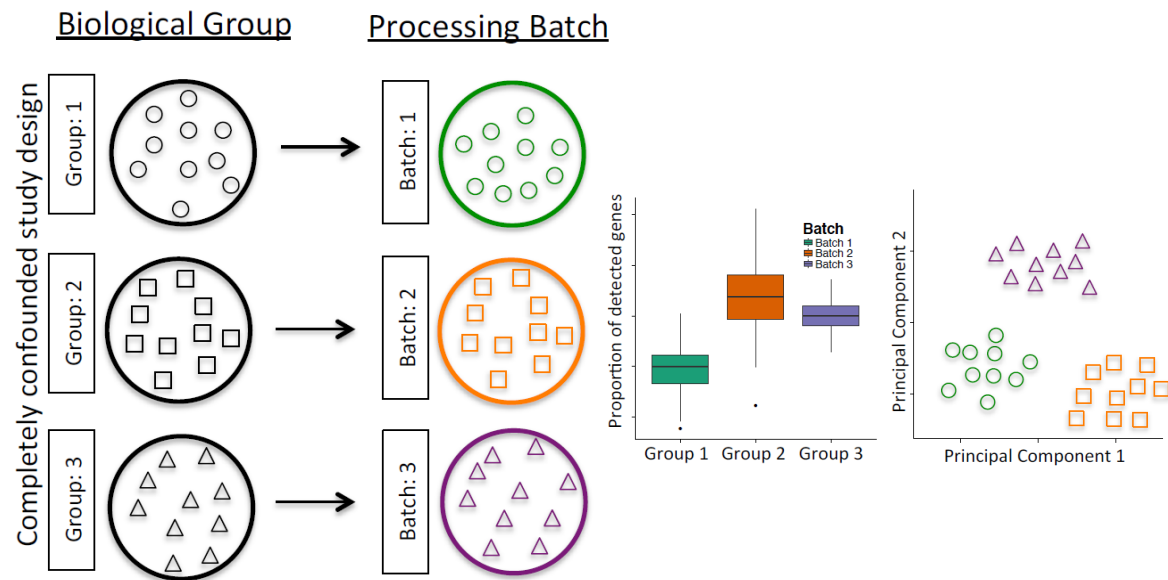


Disease-associated microglia (DAM)



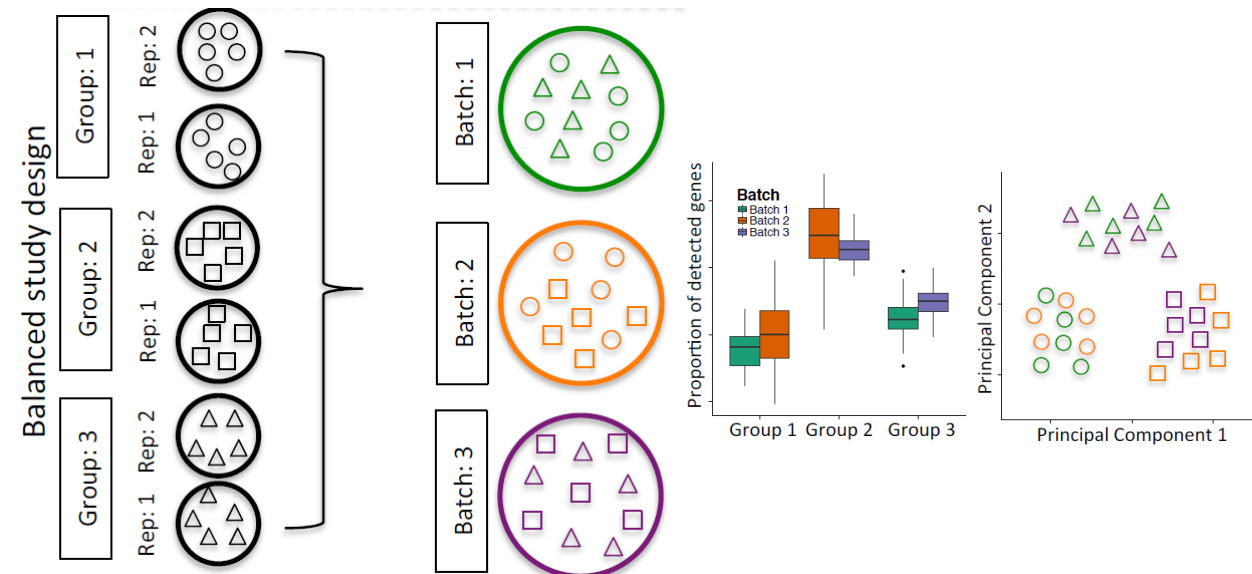
Study design

Confounded design



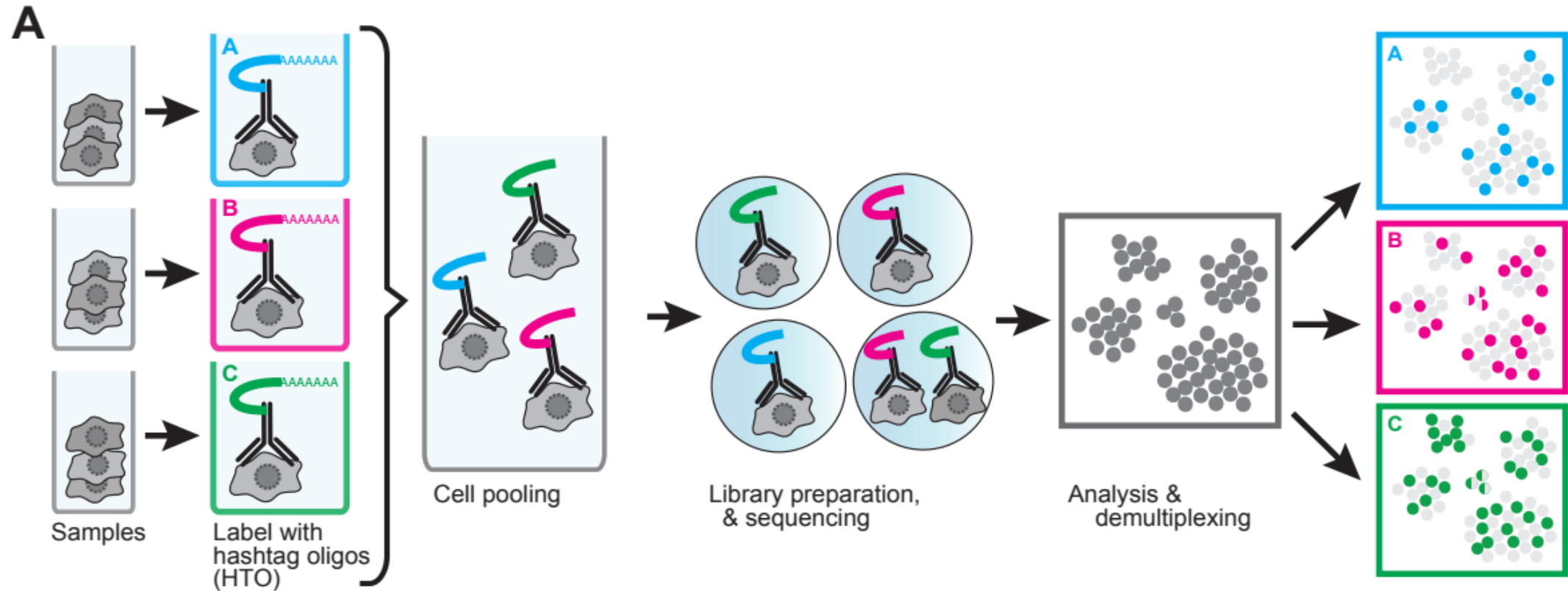
Don't design your experiment like this!!!

Not confounded design

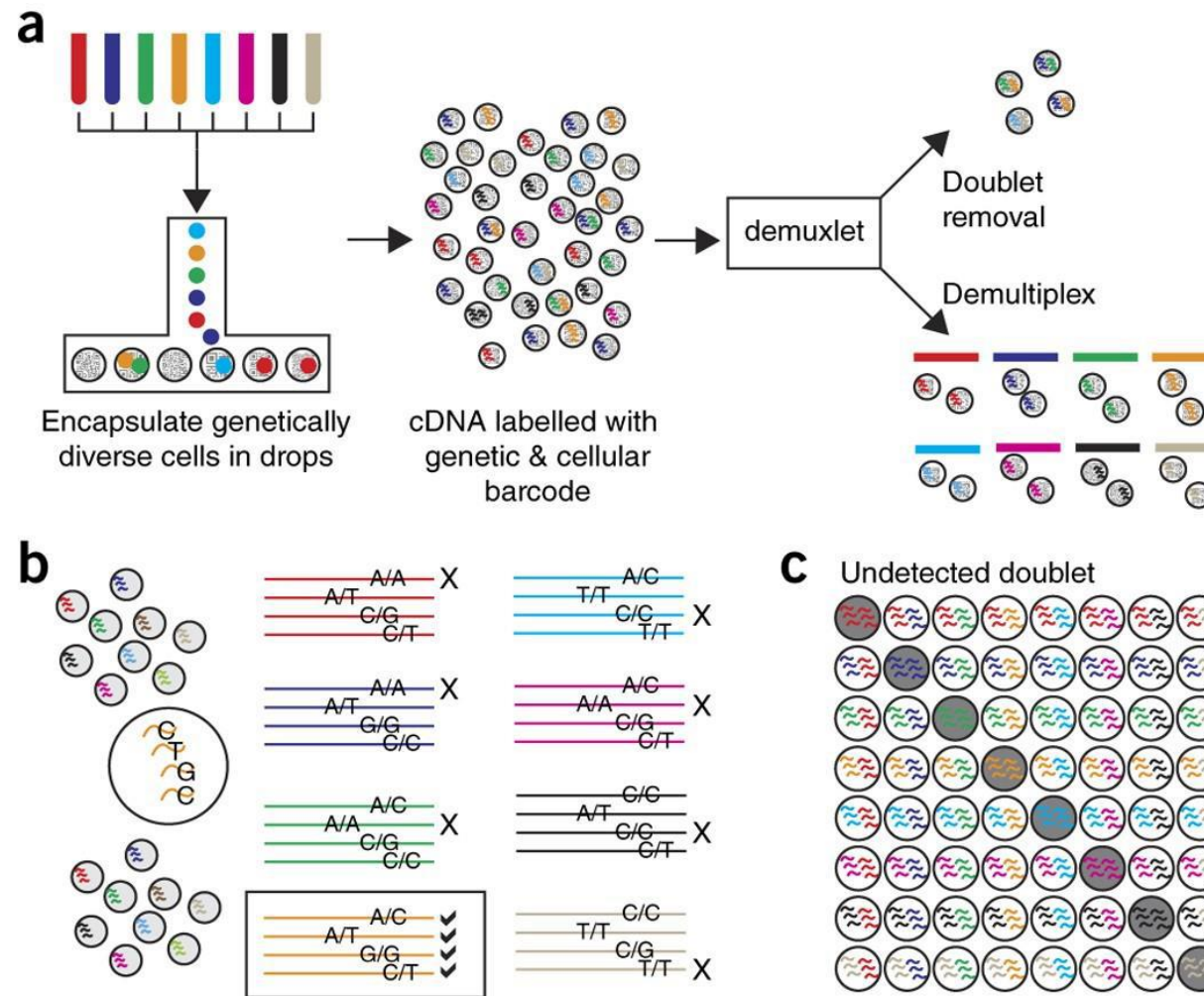


Good experimental design *does not remove batch effects*, it prevents them from biasing your results.

Cell hashing / multiplexing option 1



Cell hashing / multiplexing option 2



How many cells?

- Consider sample heterogeneity and subpopulation frequency
- Heterogenous populations require MORE cells
 - Here, a larger number of cells will provide greater statistical power
 - Heterogeneity can be reduced through experimental design (sorting, transgenics)
- Identification of rare cells requires MORE cells

How many samples?

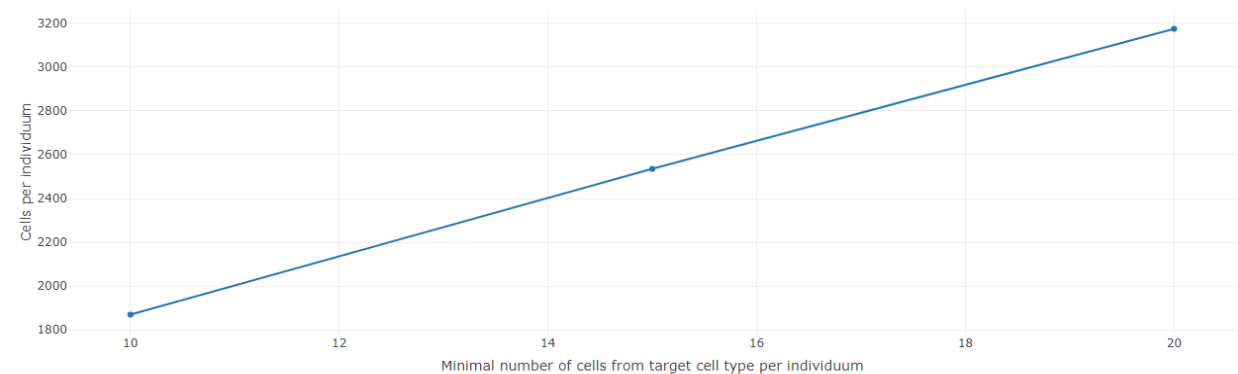
- Power calculations + group size estimates
- Is other single-cell data available (flow or mass cytometric data)?
- Computational tools available:
 - [How many cells?](#)
 - [SCOPIT](#)
 - [scPower](#)
- Pilot experiment recommended before large-scale data production

Study parameters

Samples	5
Cell type frequency	0.01
Detection power	0.95
Minimal number of cells	<input type="range" value="10 22"/>

1 11 21 31 41 51 61 71 81 91 100

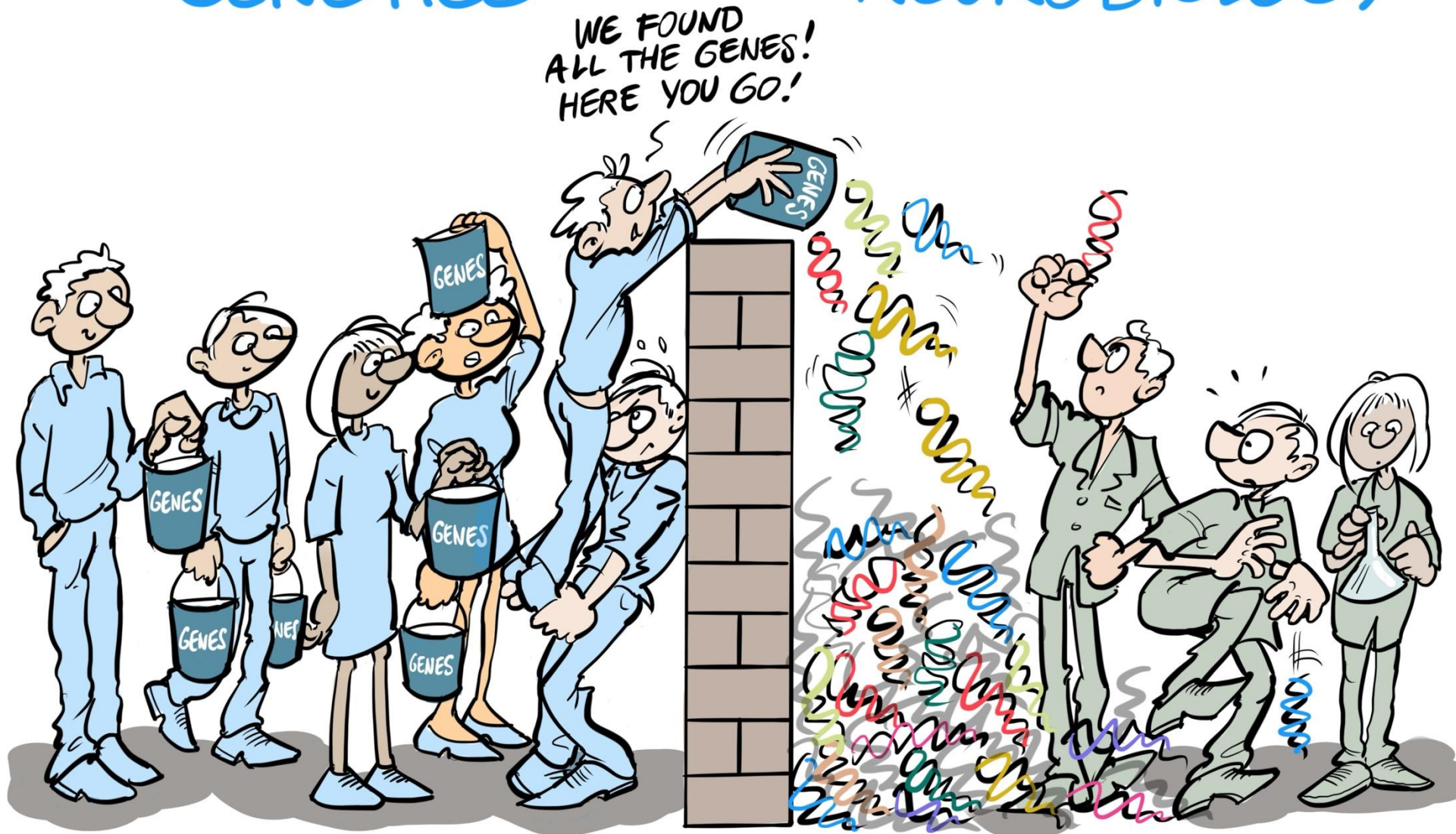
Required cells per person to detect rare cell types with a certain power



The figure shows the required number of cells per individual (y-axis, log scale) to detect the minimal number of cells from a target cell type per individual (x-axis) with a certain probability. The power depends on the total number of individuals and the frequency of the target cell type. Note that the required number of cells per sample only counts correctly measured cells (no doublets etc), so the number is a lower bound for the required cells to be sequenced.

GENETICS

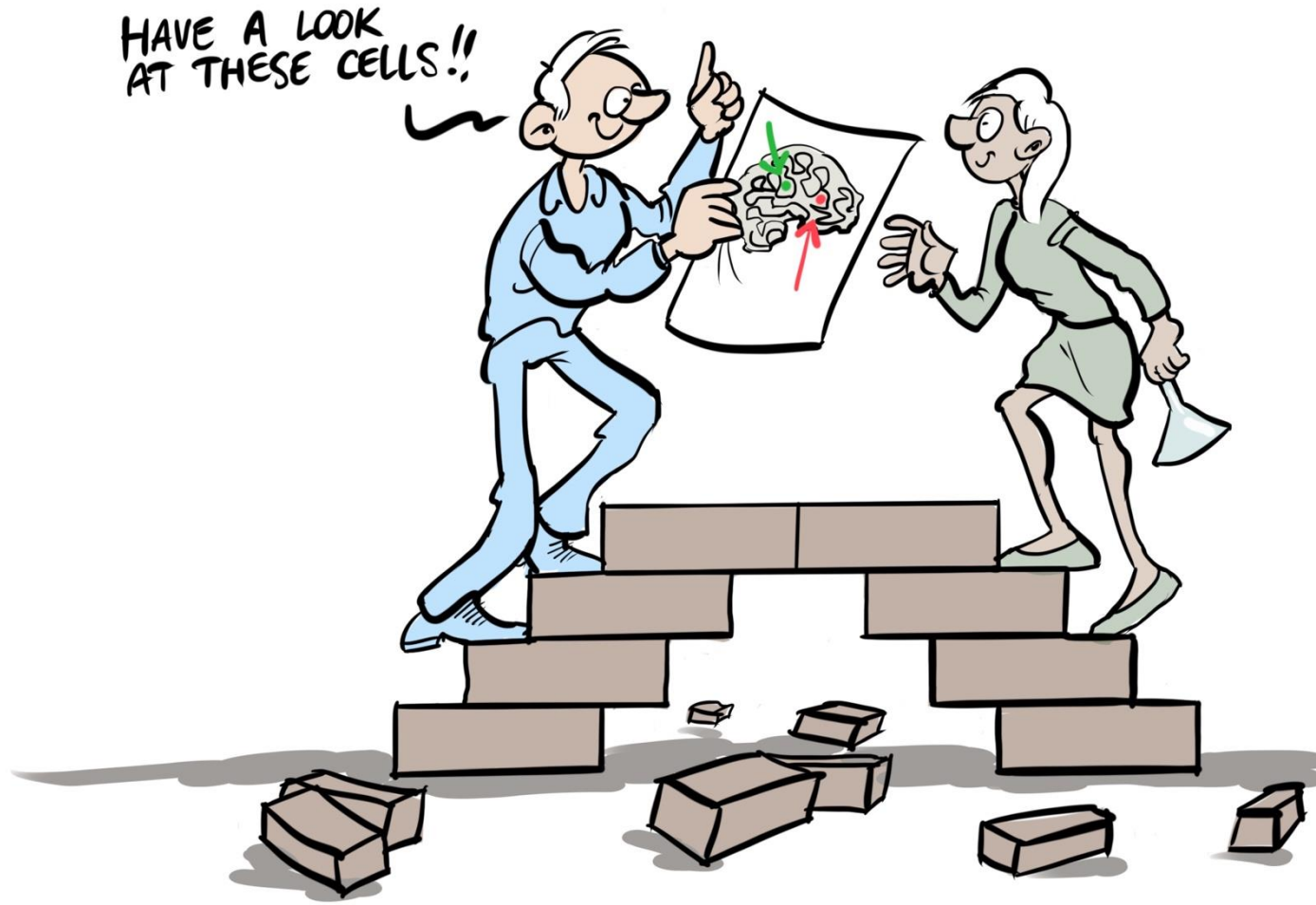
NEUROBIOLOGY



GENETICS

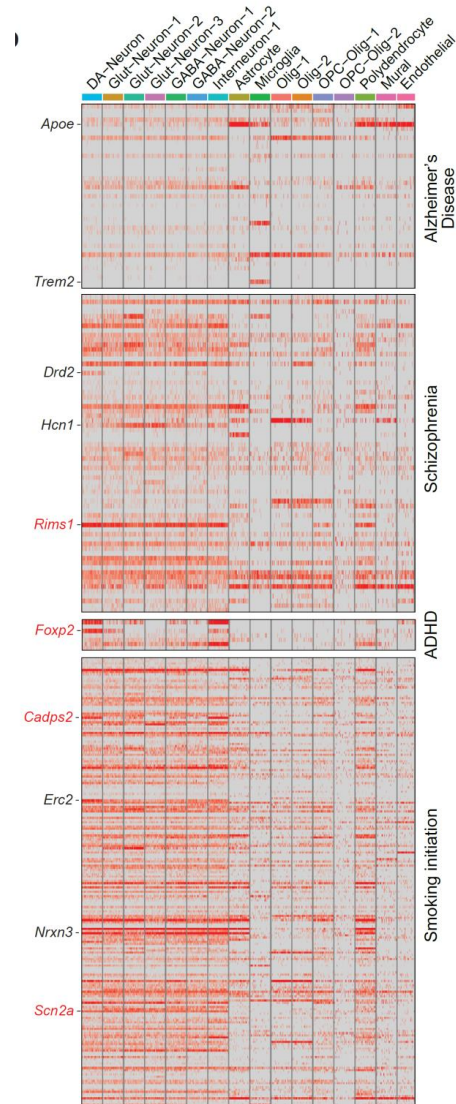
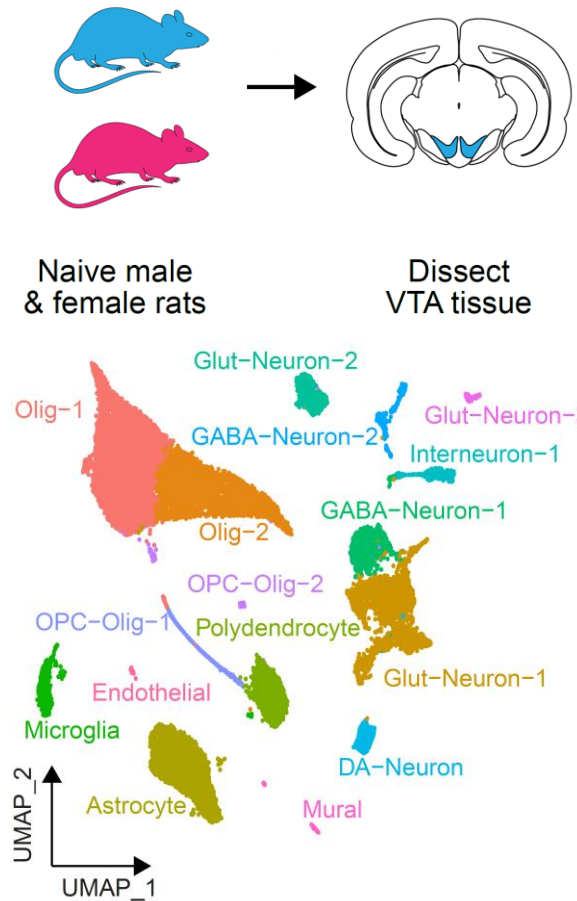
NEUROBIOLOGY

HAVE A LOOK
AT THESE CELLS!!

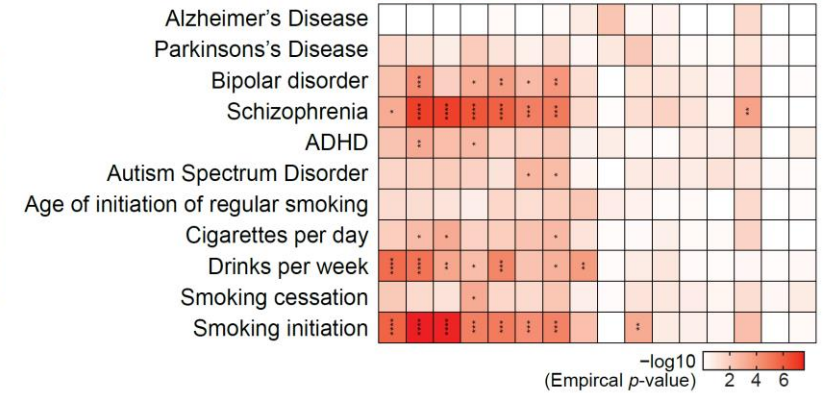


2. Cell-type specificity of diseases variants

Rat ventral tegmental area (VTA)

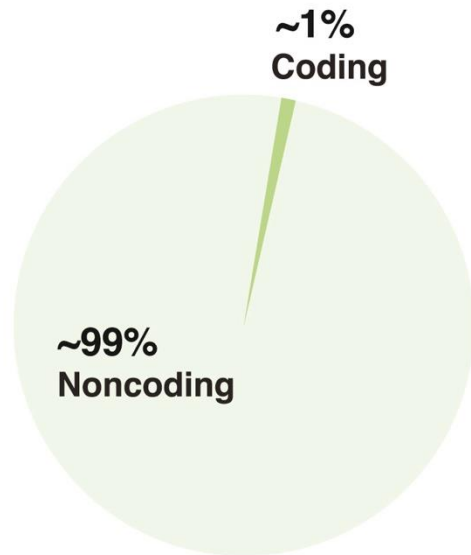


MAGMA gene set
enrichment analysis

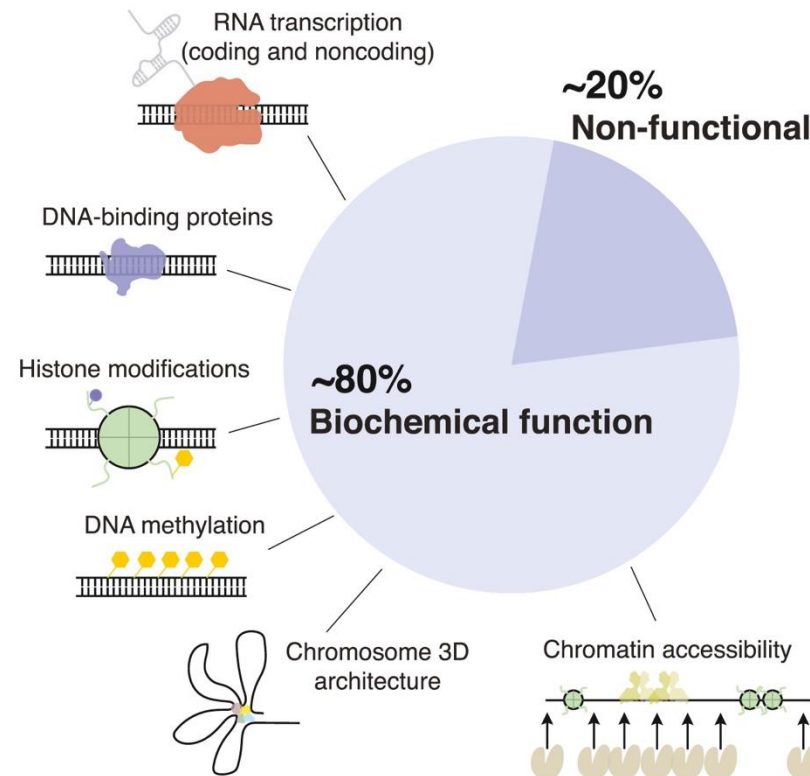


Majority of variants in non-coding regions

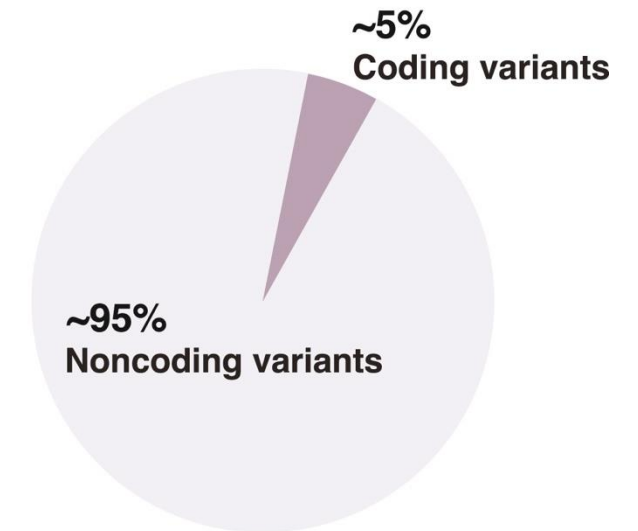
Sequence composition



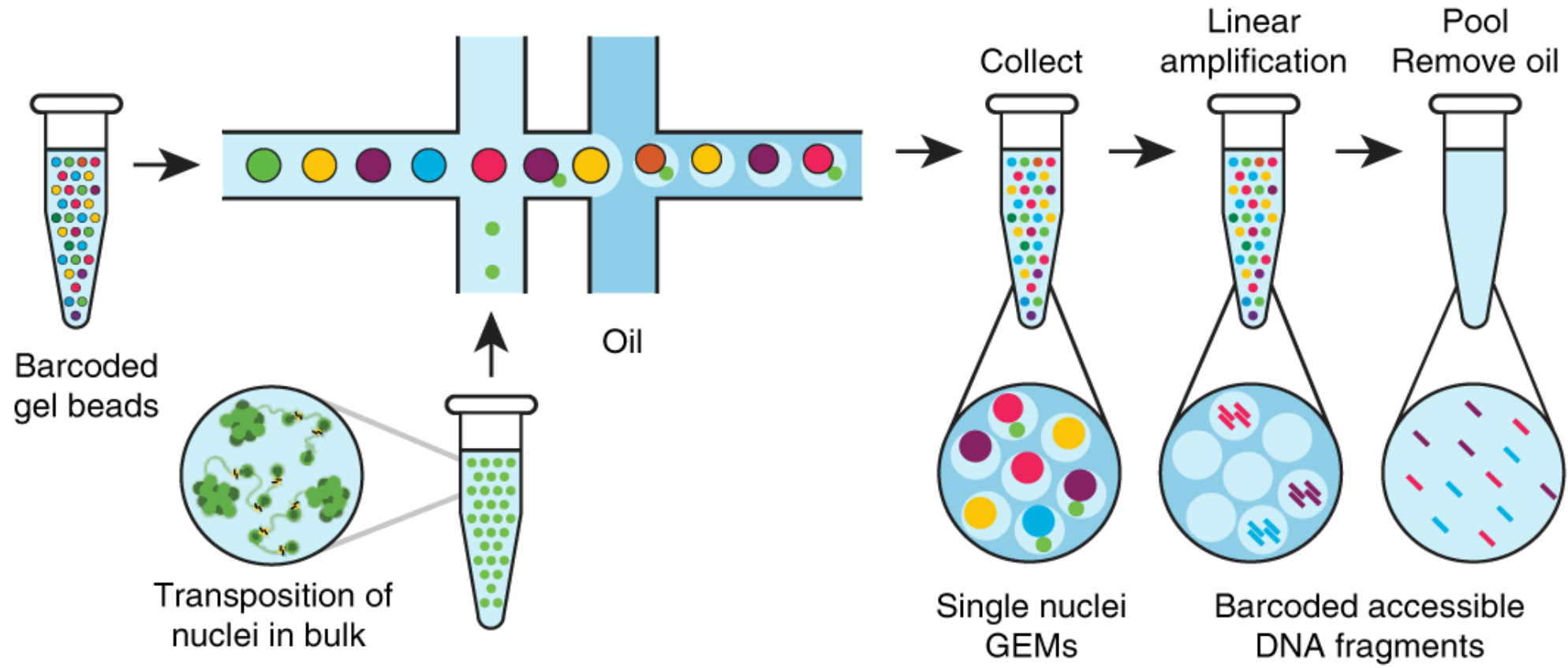
Functional mapping (ENCODE)



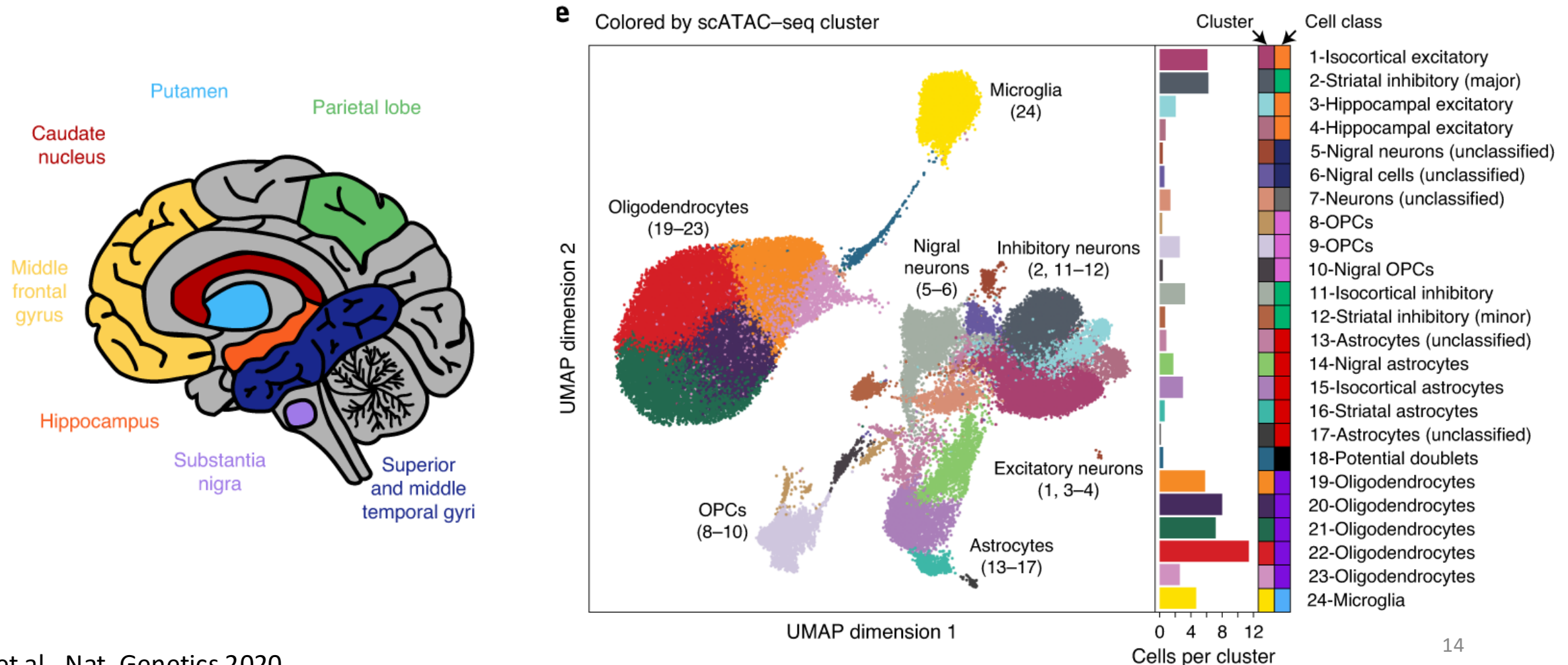
Genome-wide association studies (GWAS)



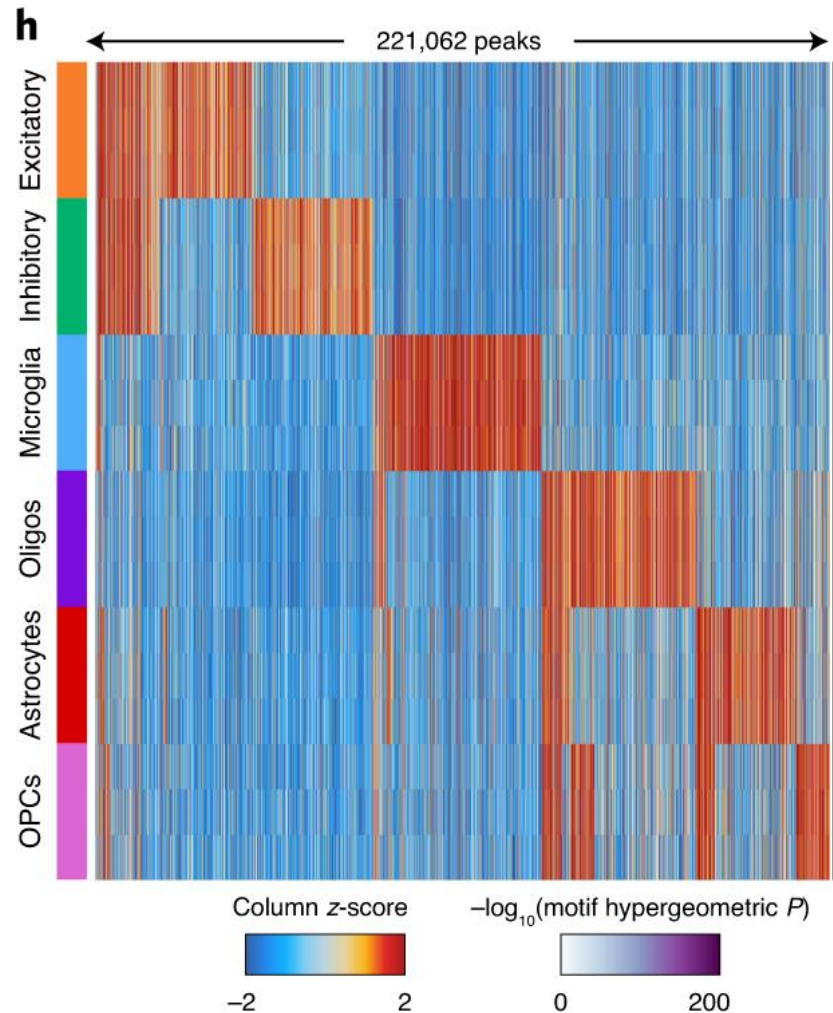
scATAC



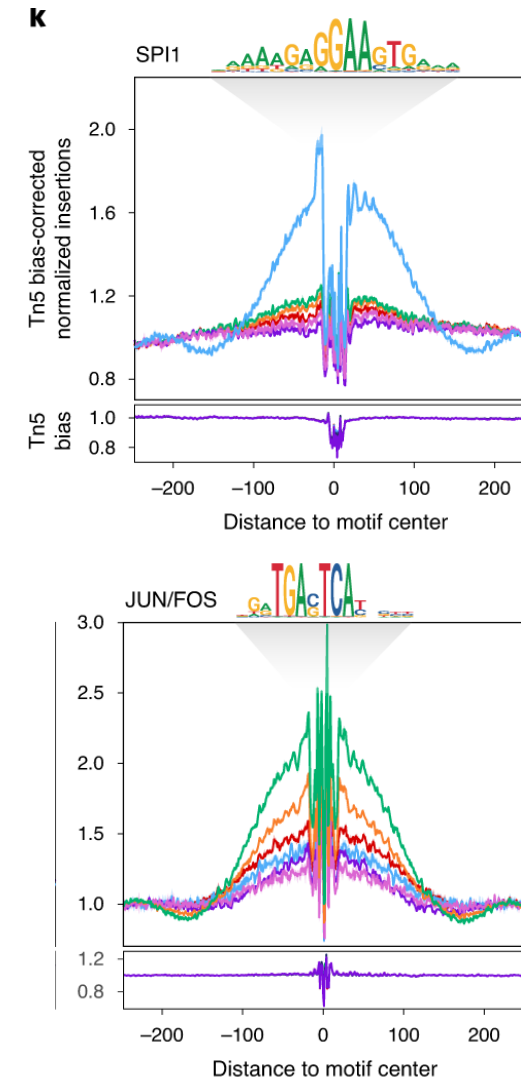
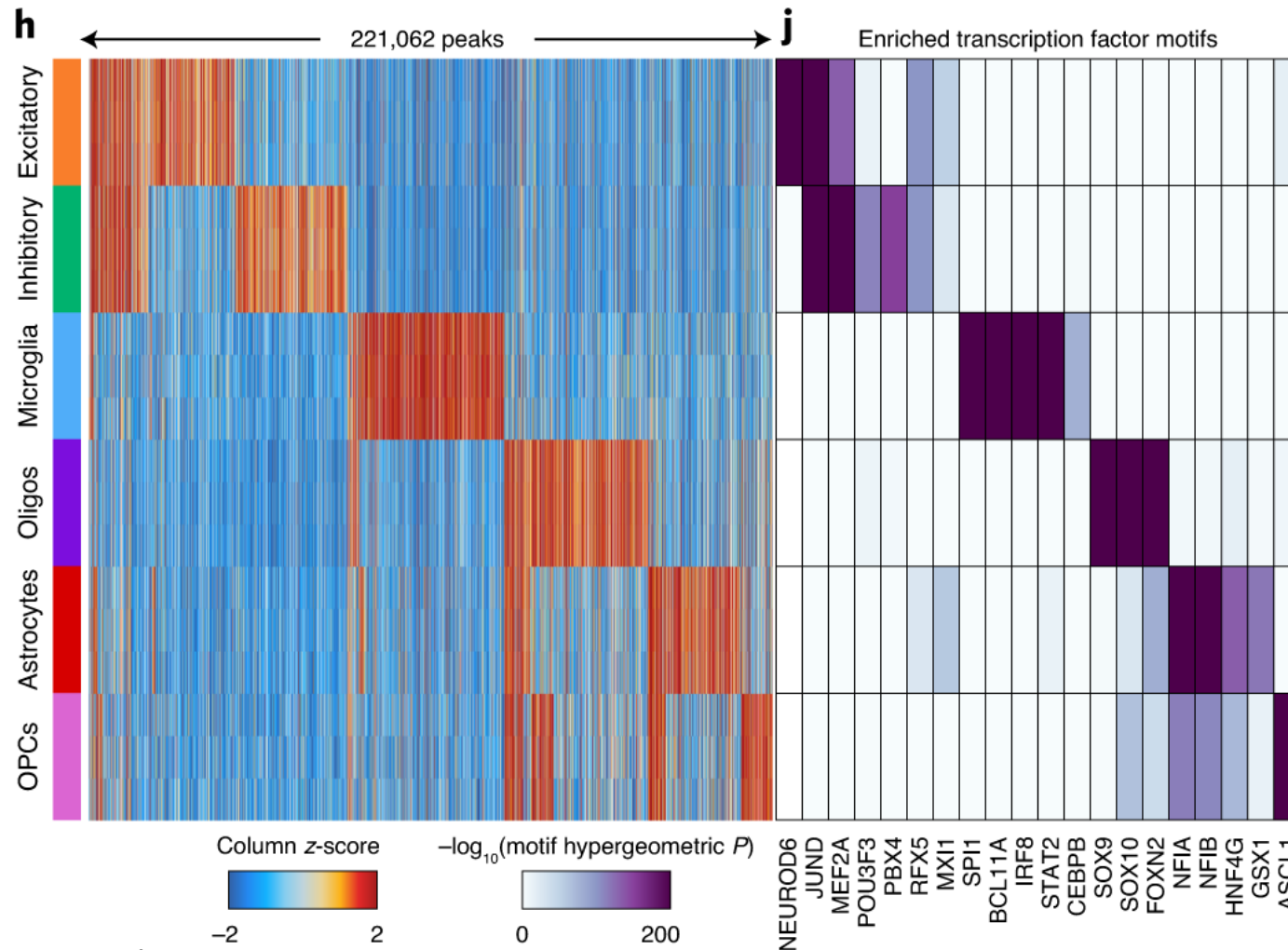
scATAC-seq pinpoints the cellular targets of GWAS polymorphisms



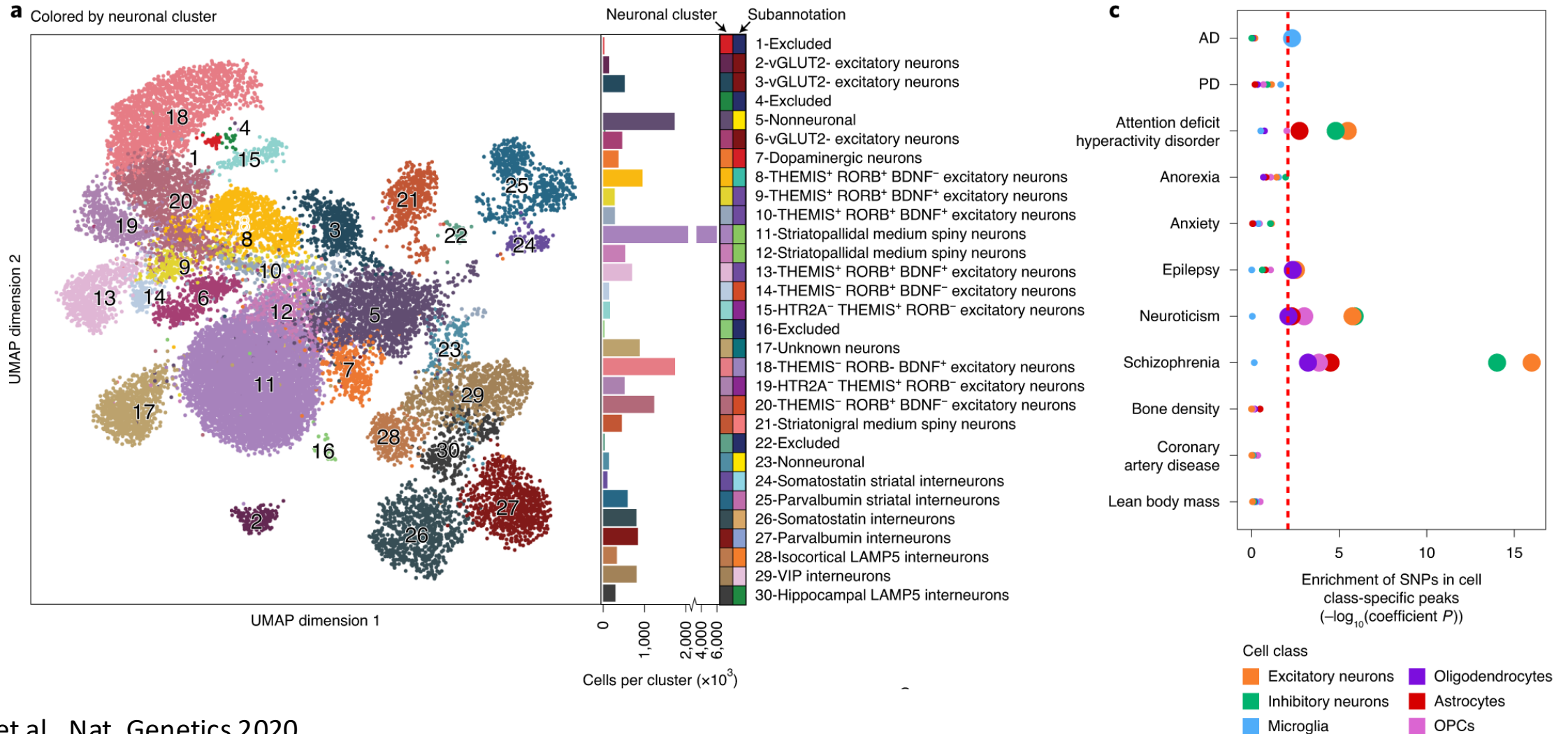
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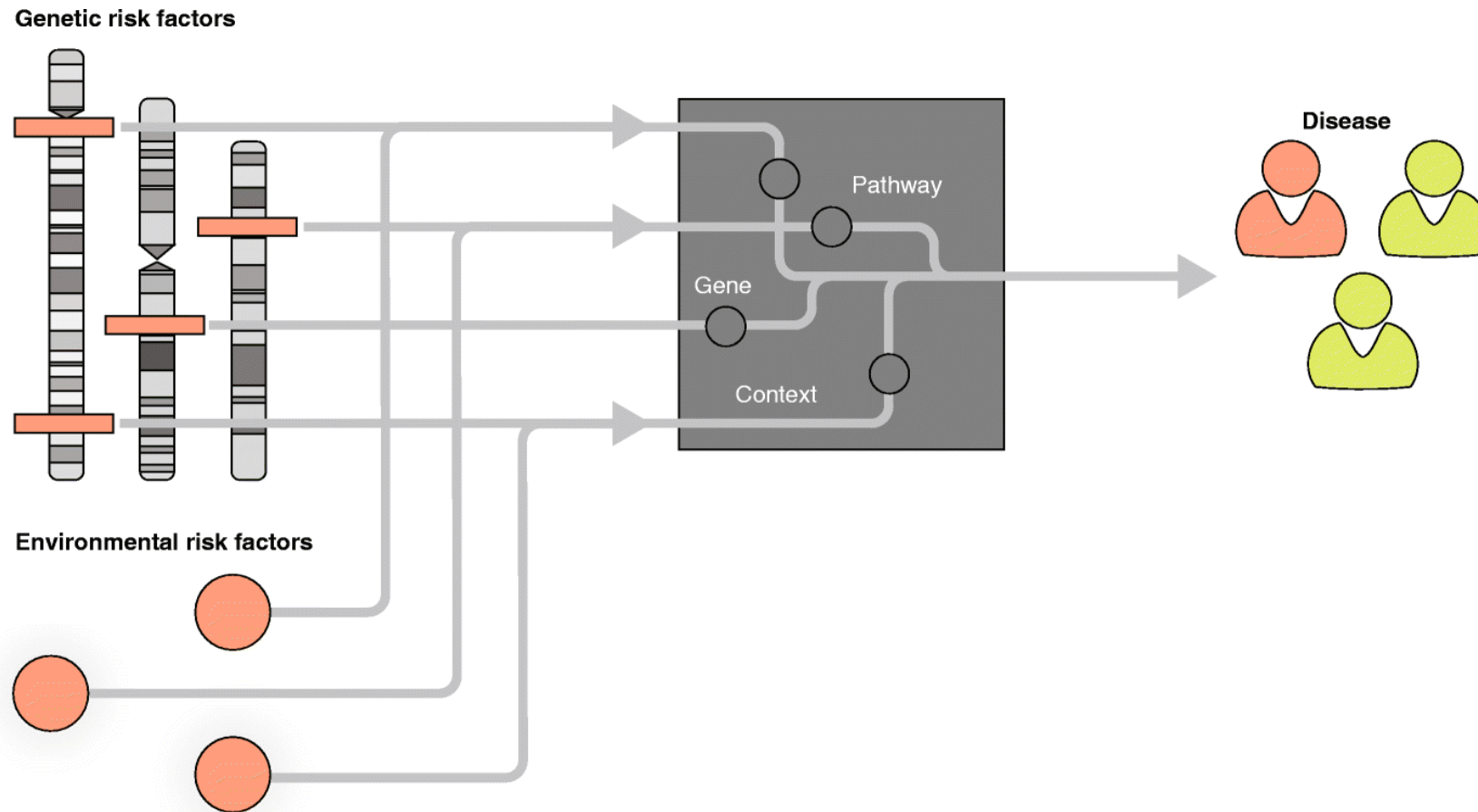
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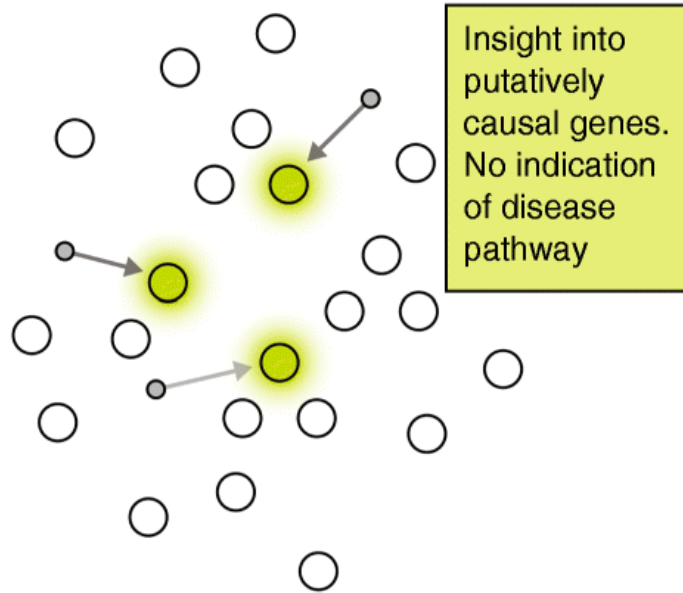


Linking genetic and environmental risk factors to disease

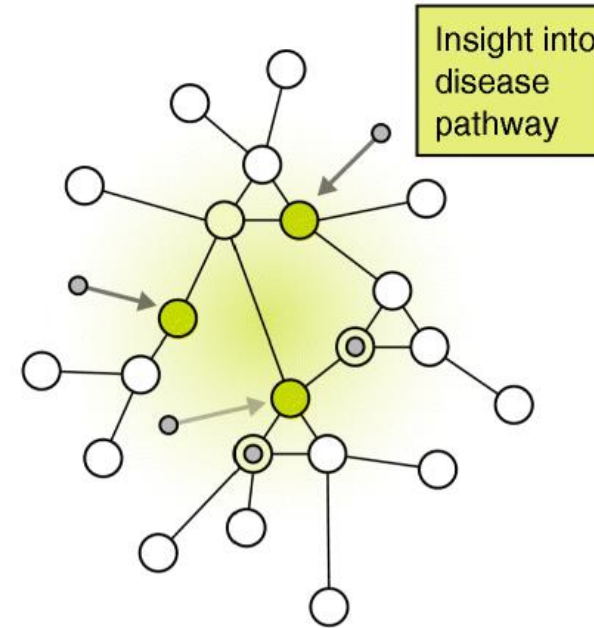


3. Expression / co-expression quantitative trait loci (eQTL)

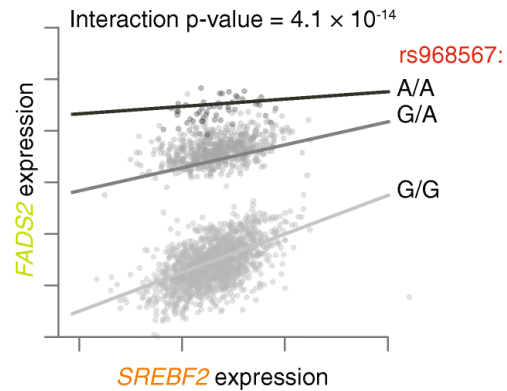
eQTL: link variants to genes



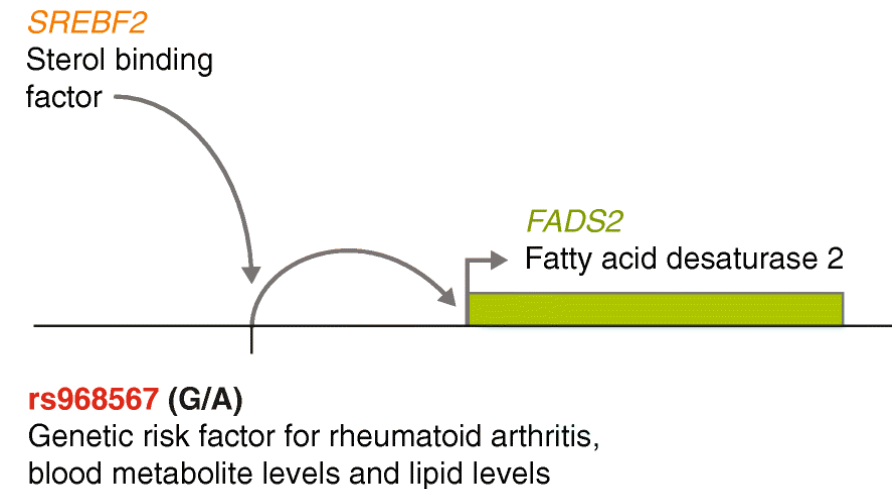
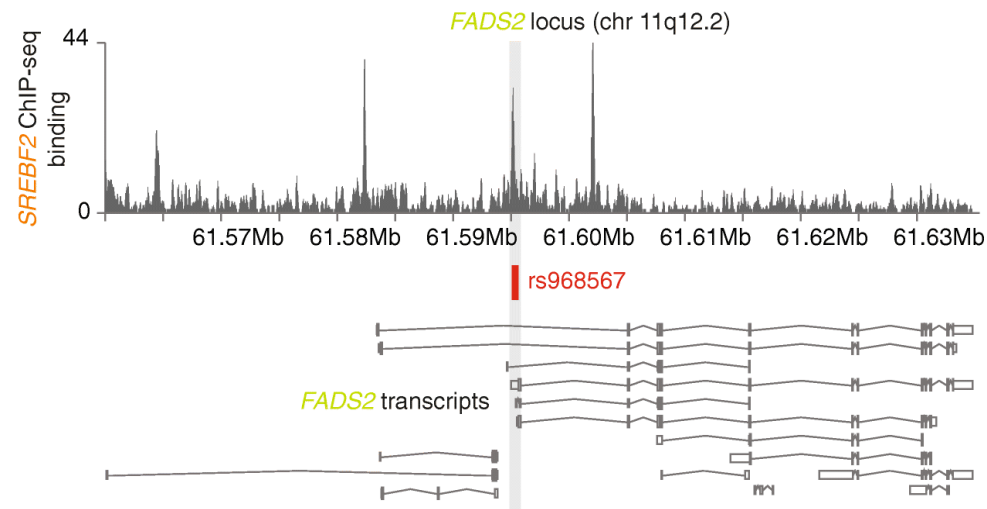
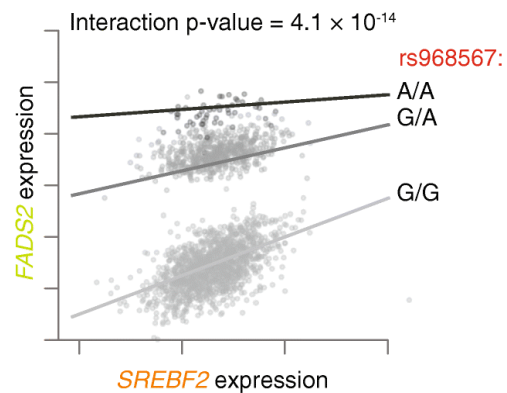
Combining eQTL and co-expression networks



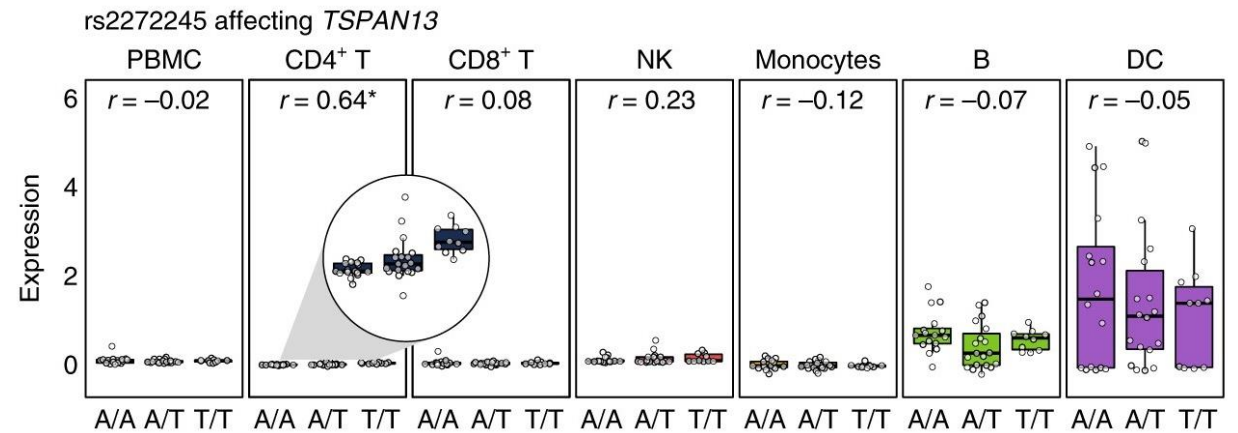
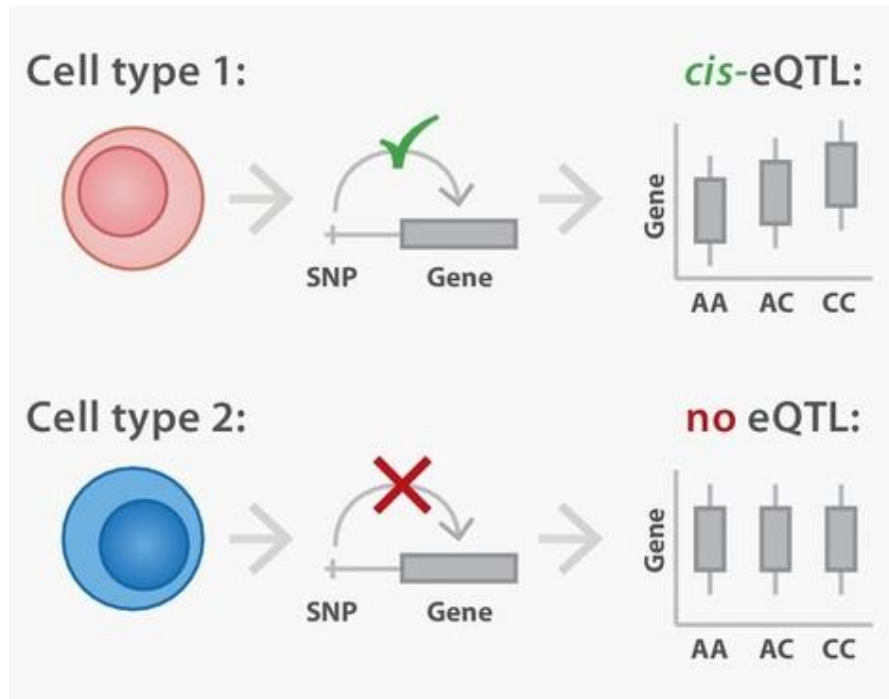
Reconstruction of a gene regulatory network using eQTLs



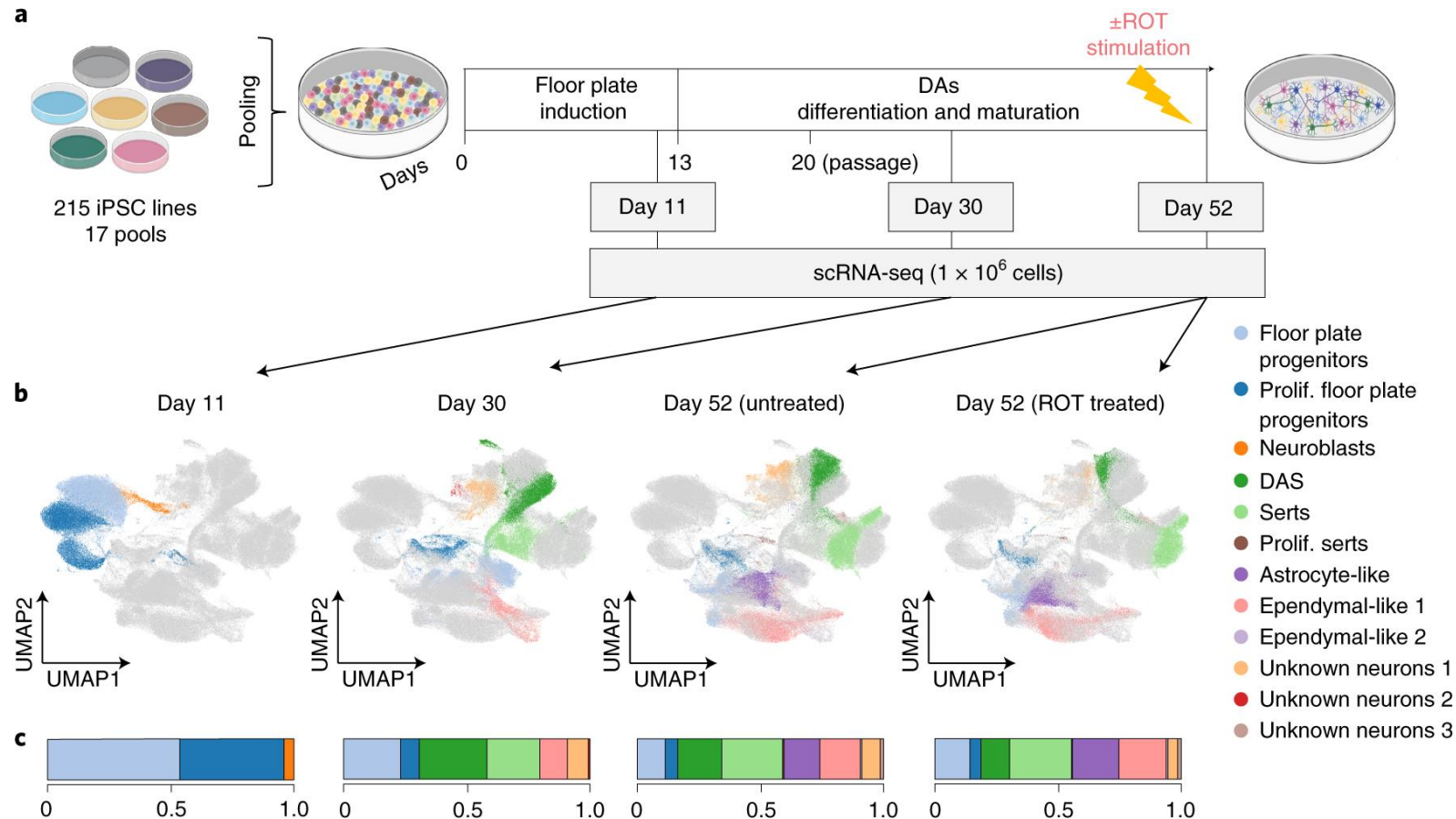
Reconstruction of a gene regulatory network using eQTLs



eQTLs are cell-type specific

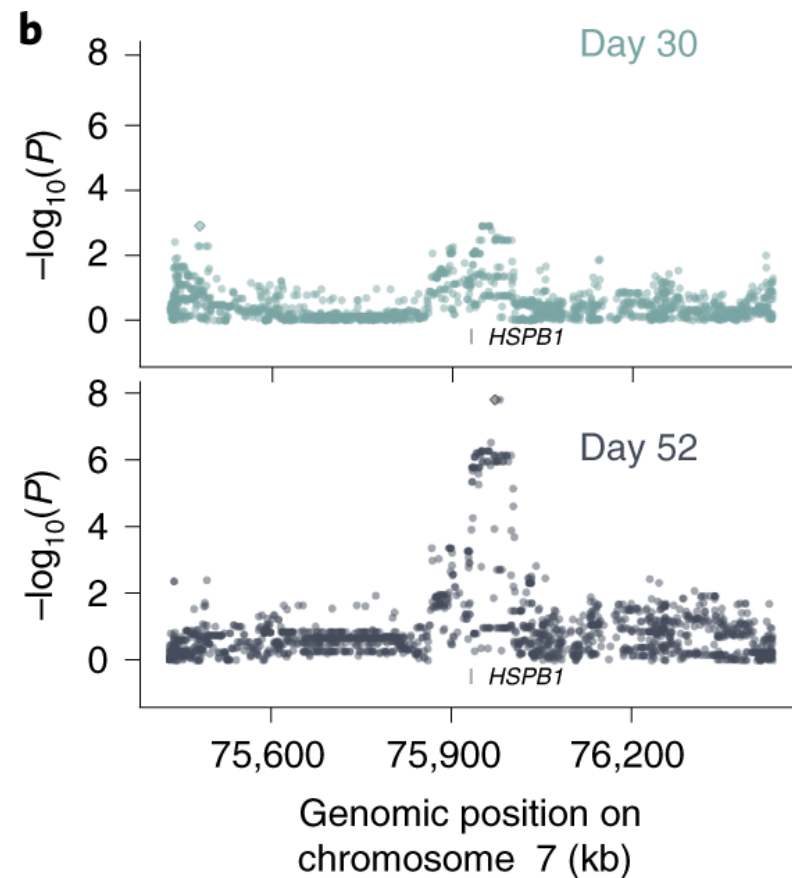


Mapping of eQTLs across midbrain differentiation

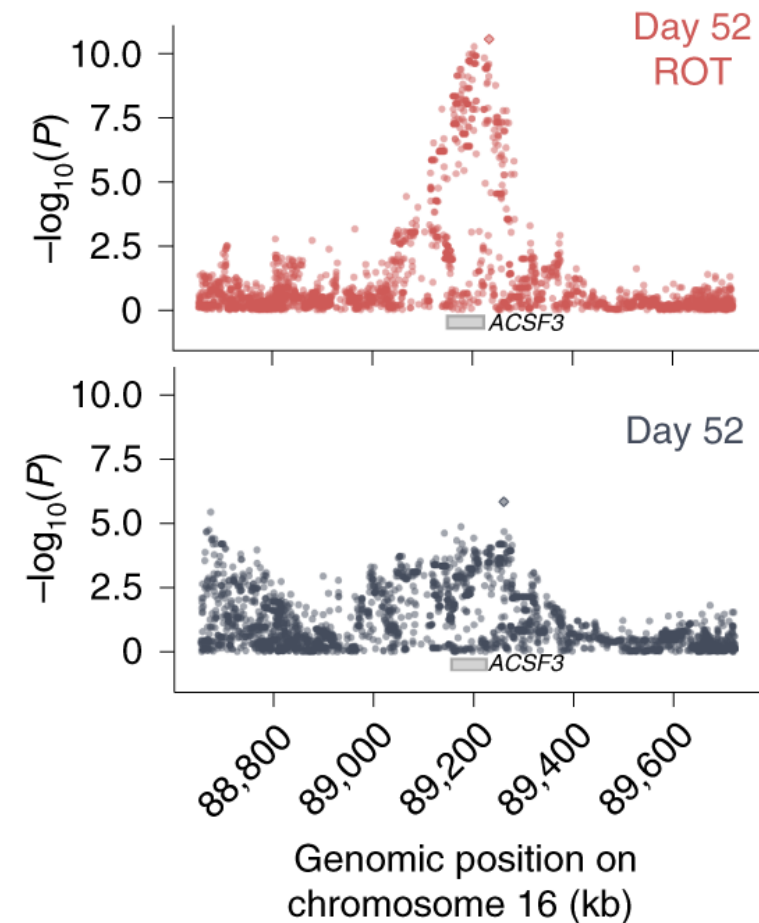


Mapping of eQTLs across midbrain differentiation

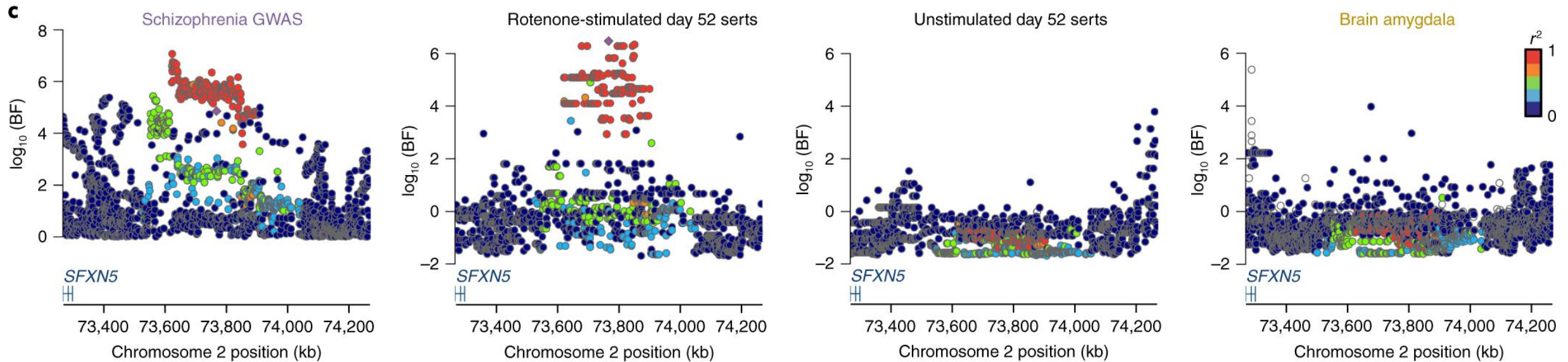
Time point-specific eQTLs in DAs



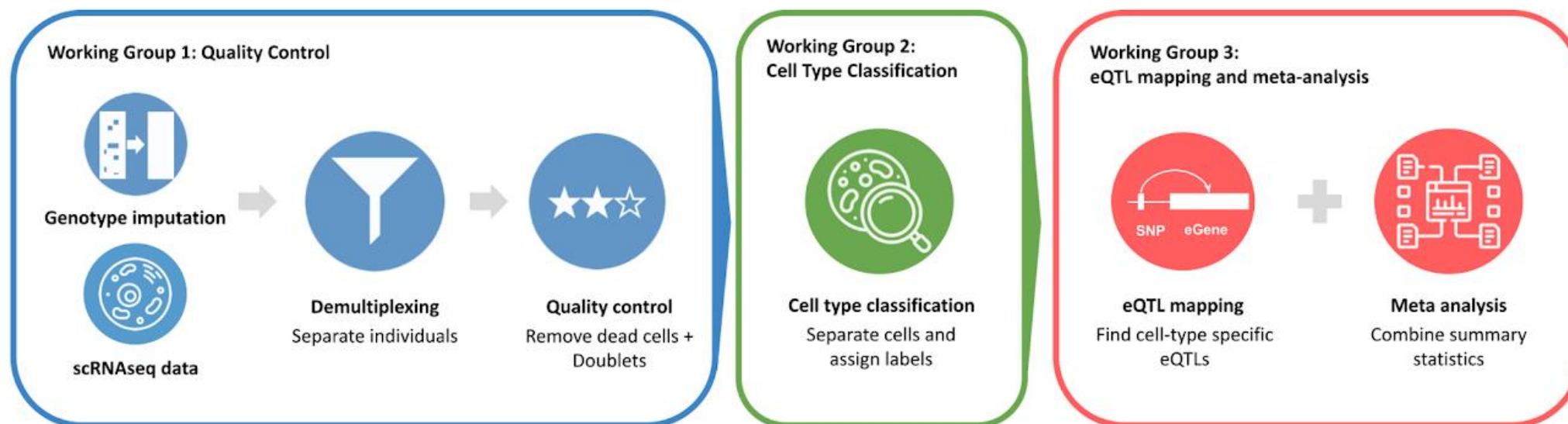
Stimulation-specific eQTLs in DAs



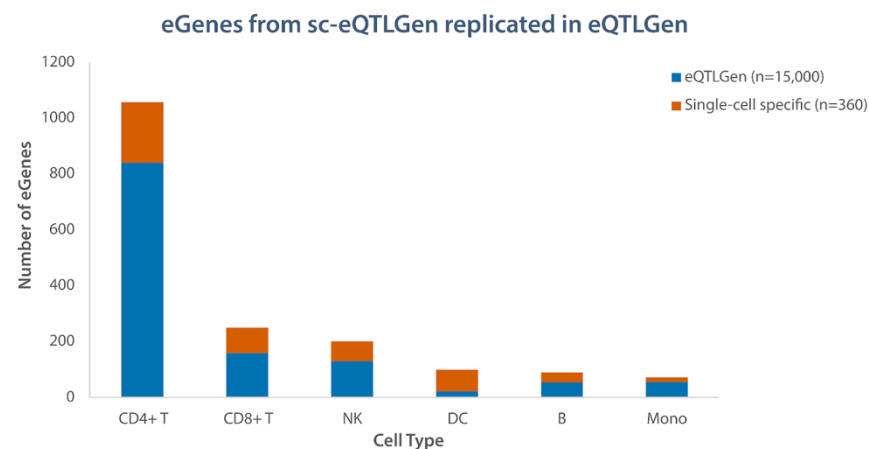
Colocalization analysis of eQTL with 25 neuro-related GWAS traits



single-cell eQTL analysis

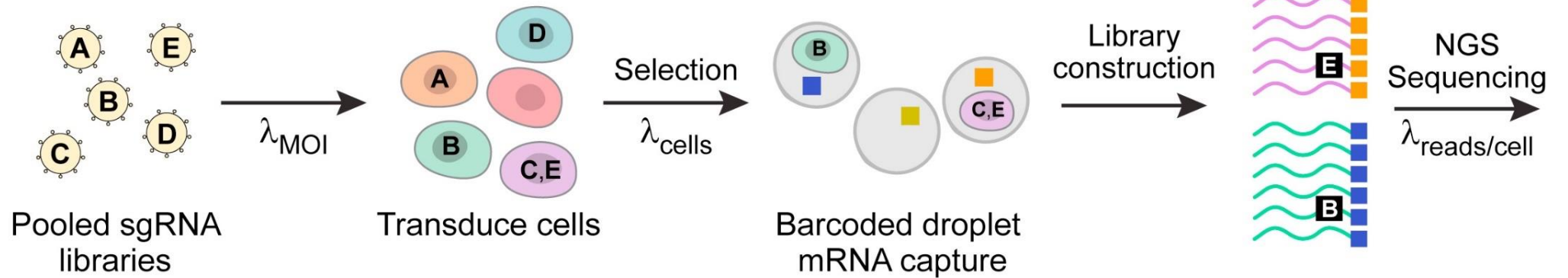
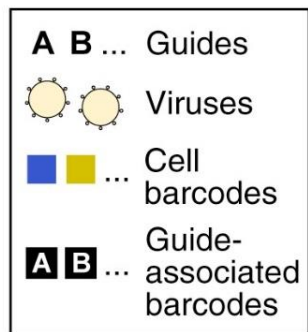


Current freeze:
16 cohorts
2,068 individuals

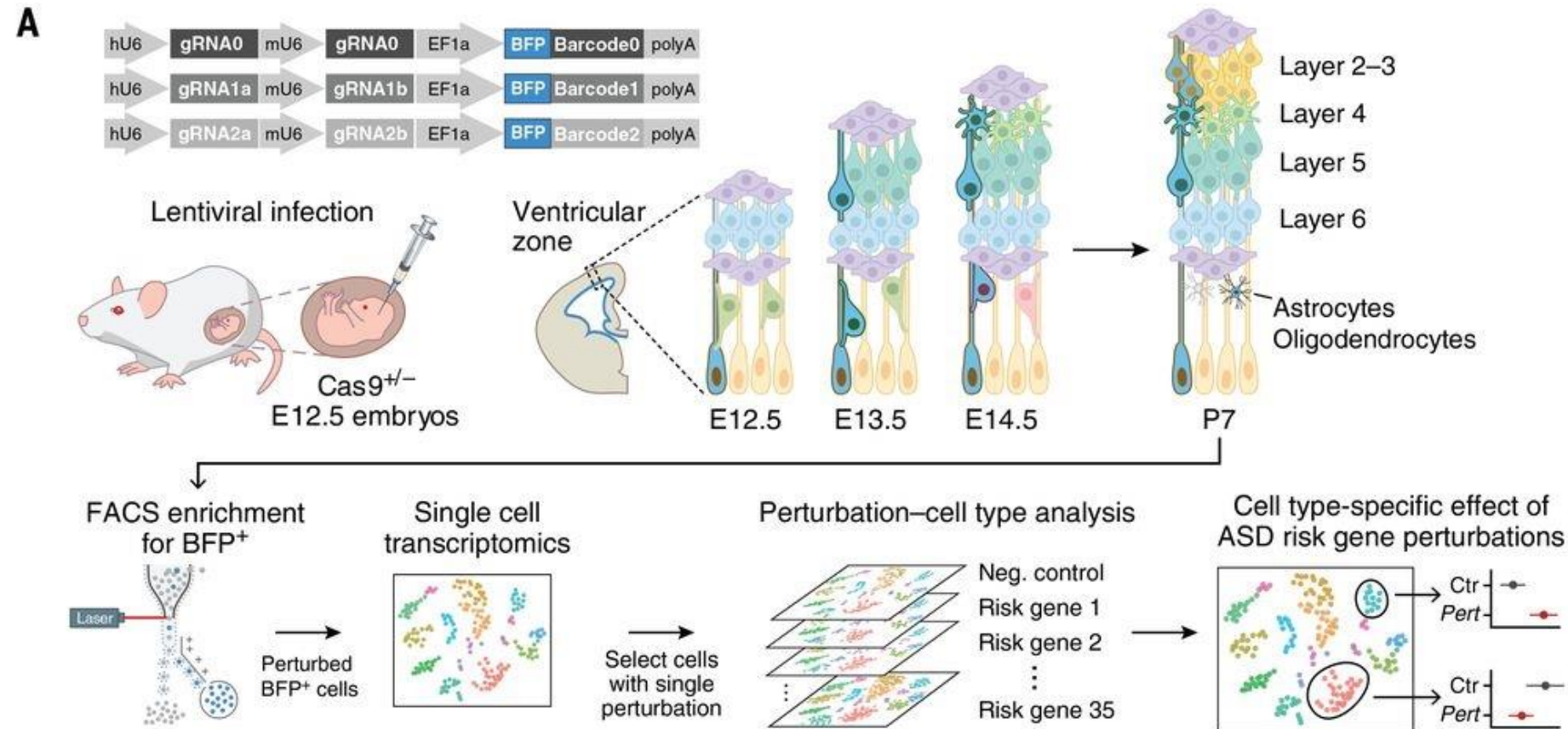


Perturb-seq

A

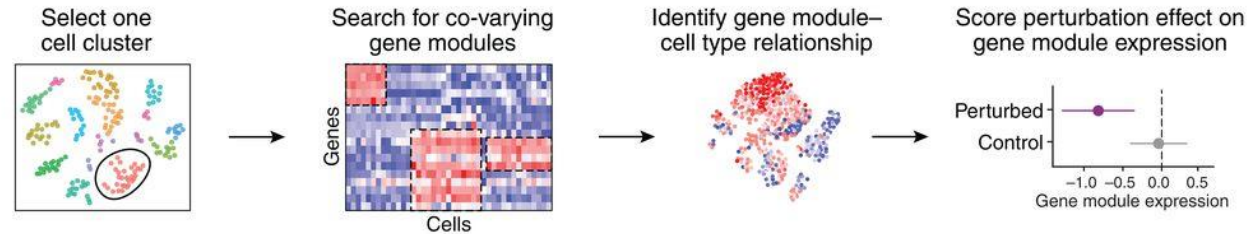


In vivo Perturb-Seq reveals neuronal and glial abnormalities associated with autism risk genes



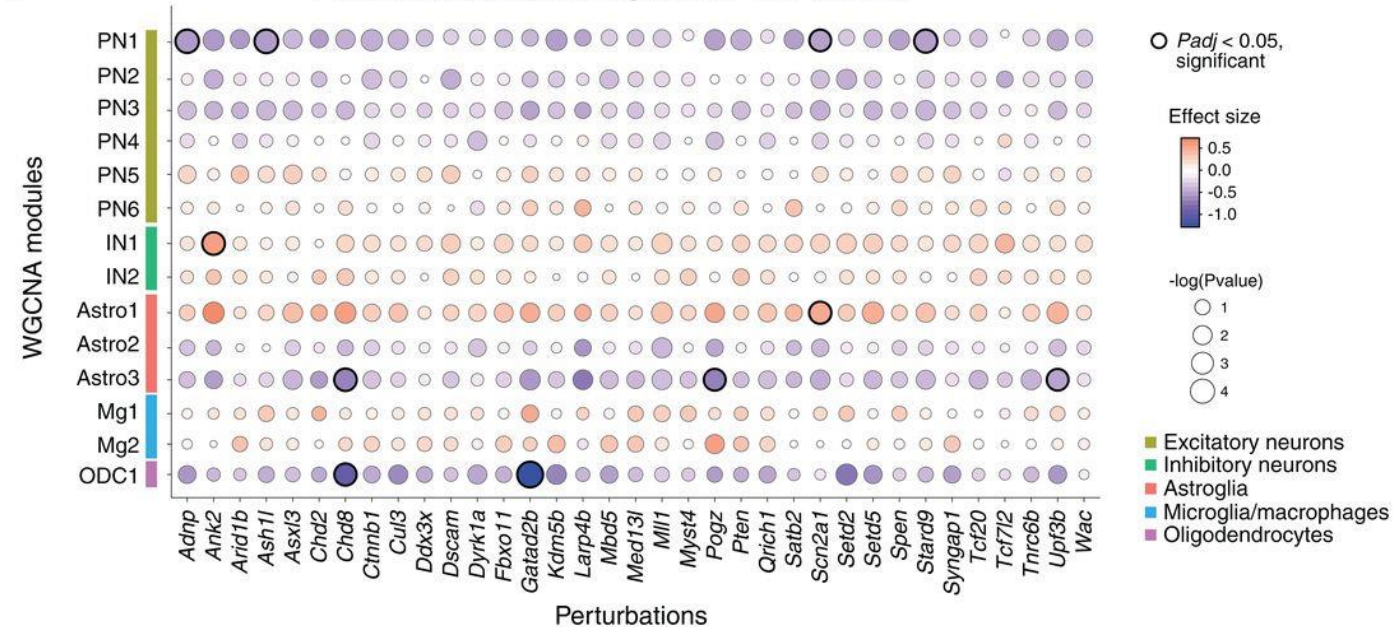
In vivo Perturb-Seq reveals neuronal and glial abnormalities associated with autism risk genes

A 14 co-varying gene modules from 5 major cell types



B

Perturbation effect compared to GFP control



Summary 3

- Single-cell data can be very helpful to pinpoint cell type context of genetic variation
- Enrichment analysis can make use of “healthy” data
- Expression and co-expression QTLs can identify cell-specific regulatory networks
- Perturb-seq allows large-scale variant screening